

AI.SPIRE

Welcome!



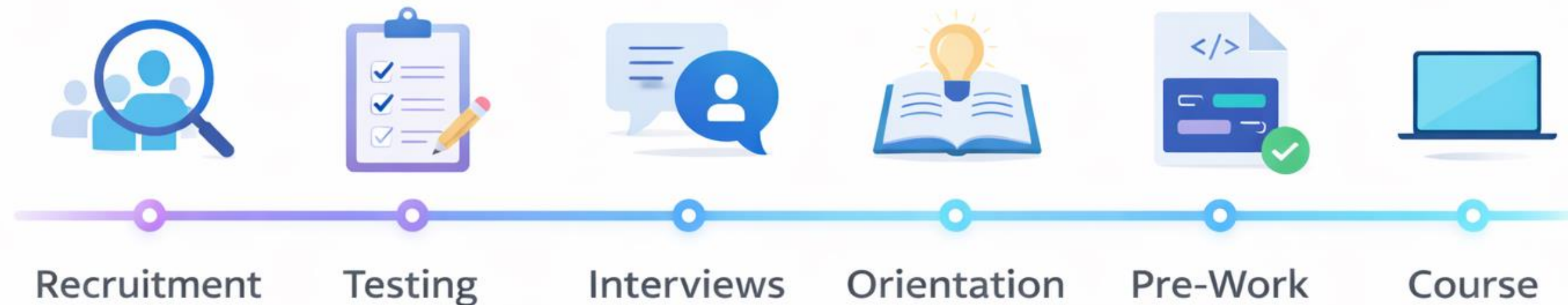
HI, I AM LANA ZUMBRUNN



- 10+ years leading global technology and strategy teams
- Worked with 200+ startup founders
- AI Engineer and Educator

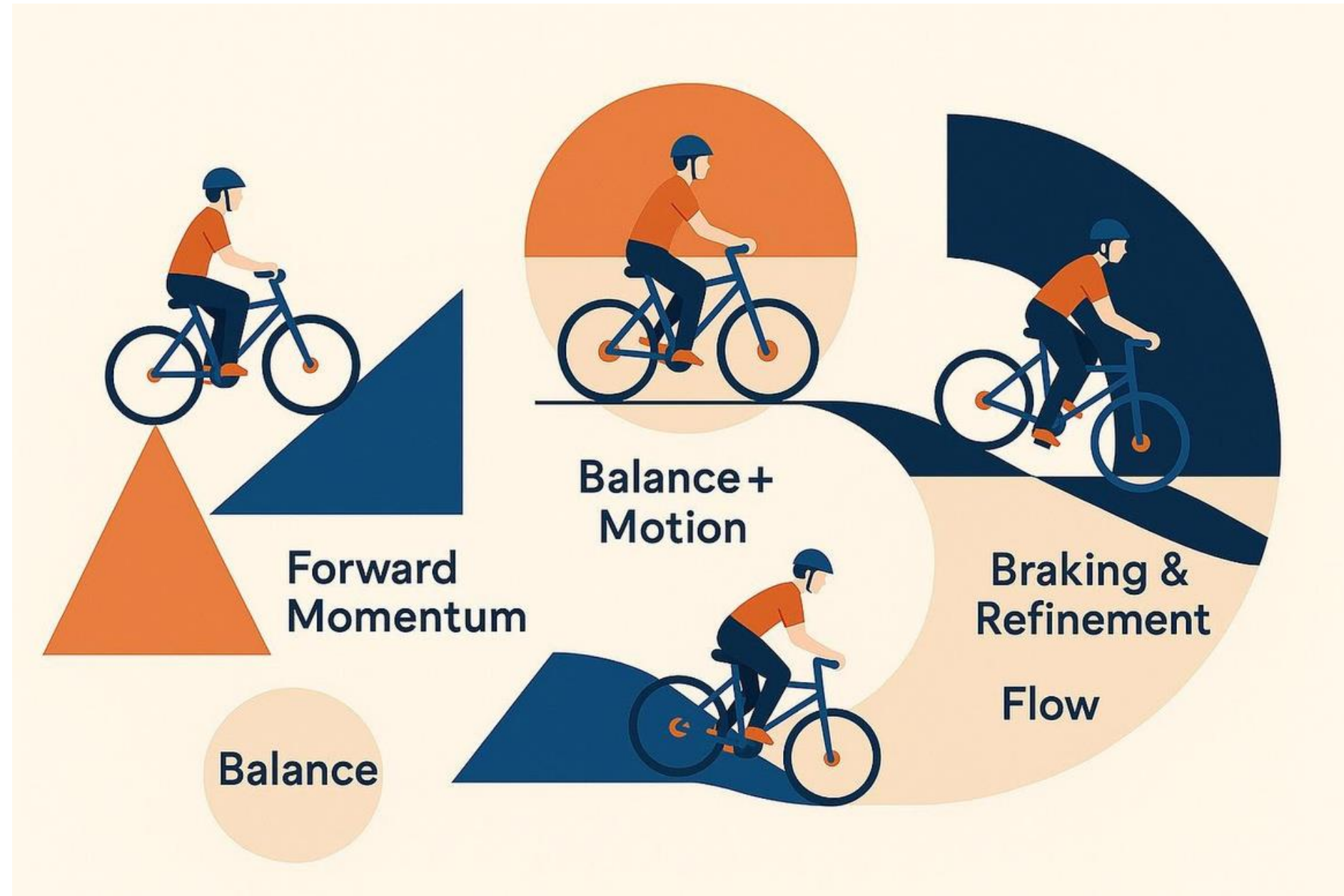
AI.SPIRE—APPLIED AI AND ML SYSTEMS

You made it!
Let's go!



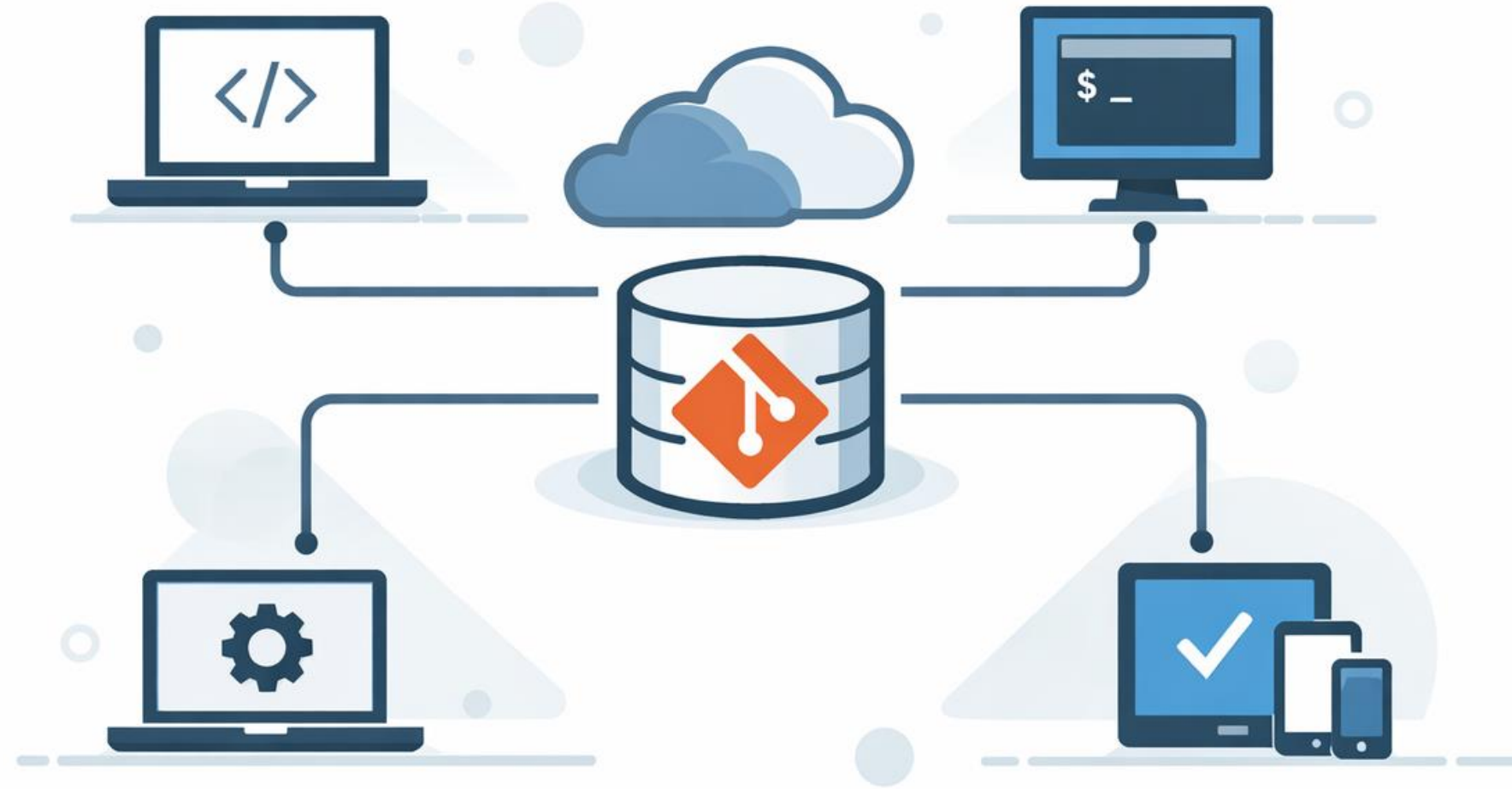
THE BUILD CYCLE

- Challenge
- Repetition
- Practice
- Reflection

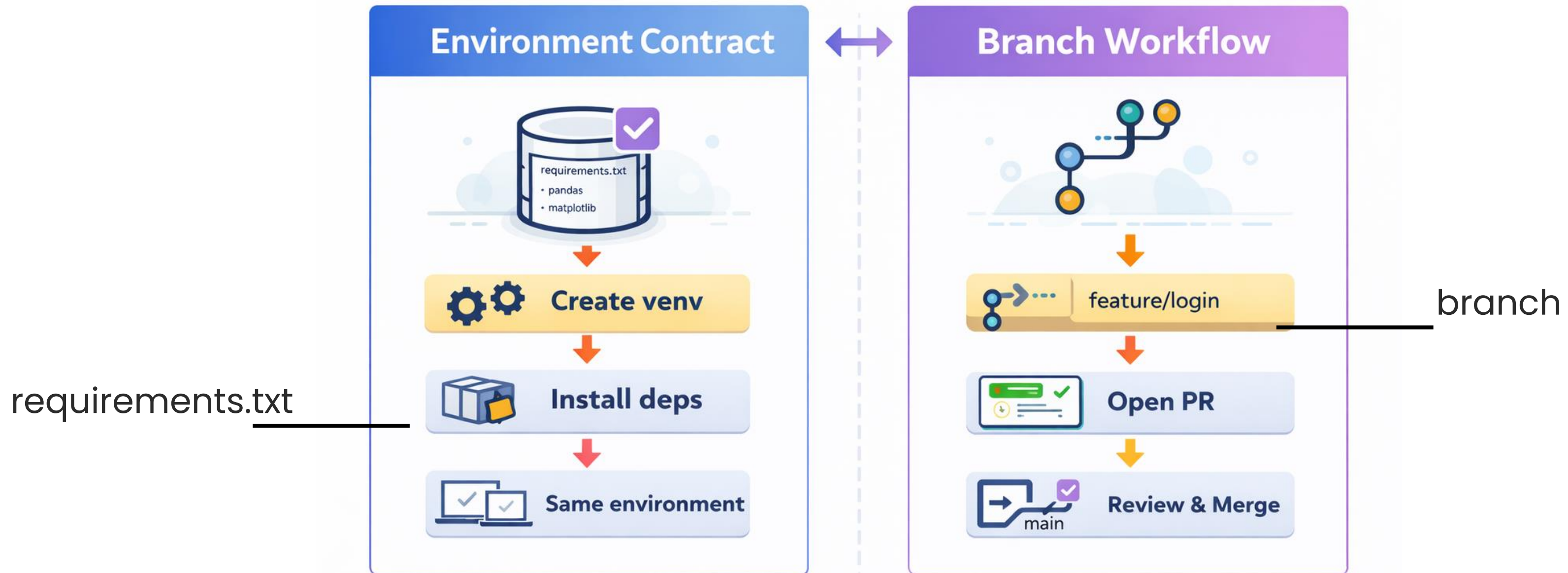


AI.SPIRE—APPLIED AI AND ML SYSTEMS

MODULE 1 — DEV ENVIRONMENT & COLLABORATION

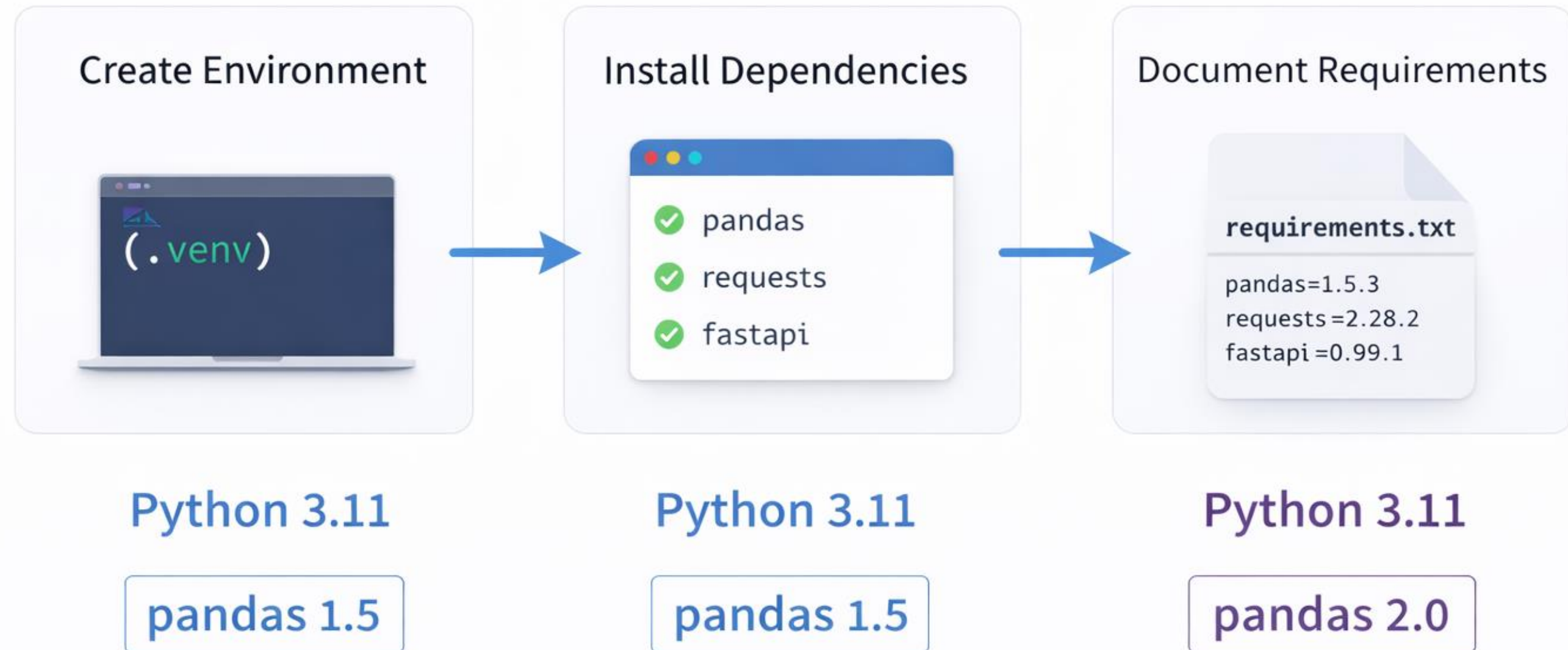


2 MENTAL MODELS



ENVIRONMENTS AS CONTRACTS: REPRODUCIBLE ENVIRONMENTS

QUESTIONS?

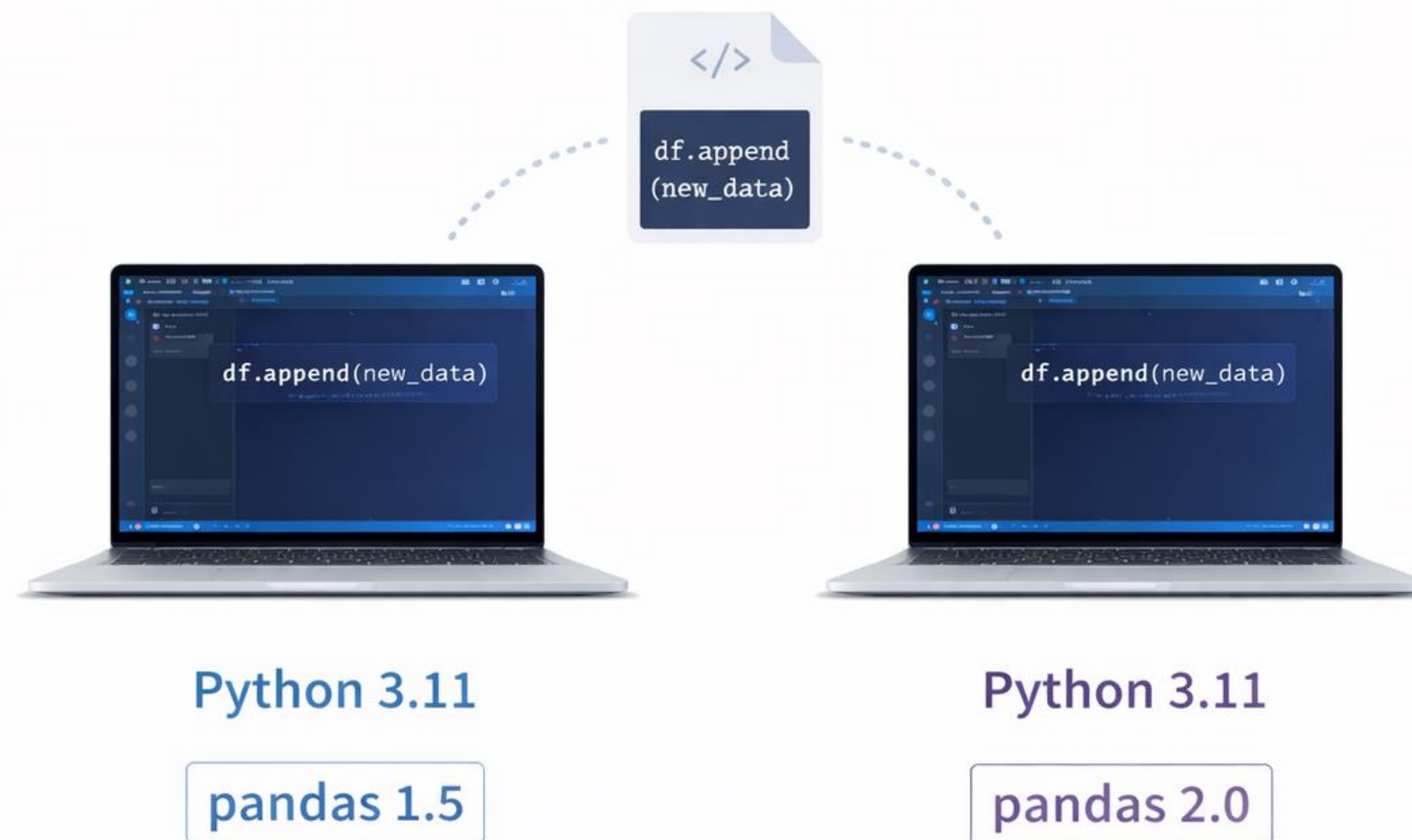


THE SETUP PROBLEM

"IT WORKS ON MY MACHINE"



SAME CODE. DIFFERENT ENVIRONMENT.



- Different Python versions
- Different library versions
- Different system dependencies

YOUR PROJECT WORKSPACE

```
(.venv) python -m venv .venv  
$ source .venv/bin/activate  
$ |
```

THE INGREDIENT LIST



Recipe

- 2 cups flour
- 1 cup sugar
- 3 eggs

=

requirements.txt

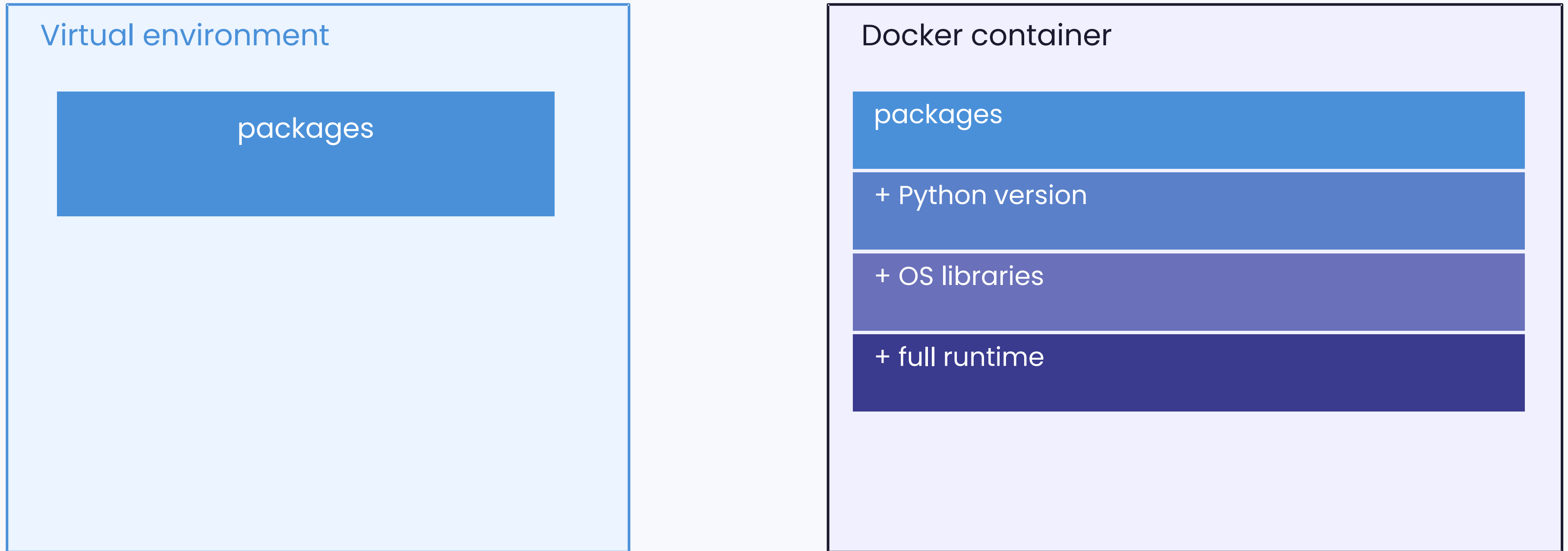
```
pandas==2.1.0  
numpy==1.26.2  
matplotlib==3.7.3
```

```
$ pip freeze > requirements.txt
```

```
$ pip install -r requirements.txt
```

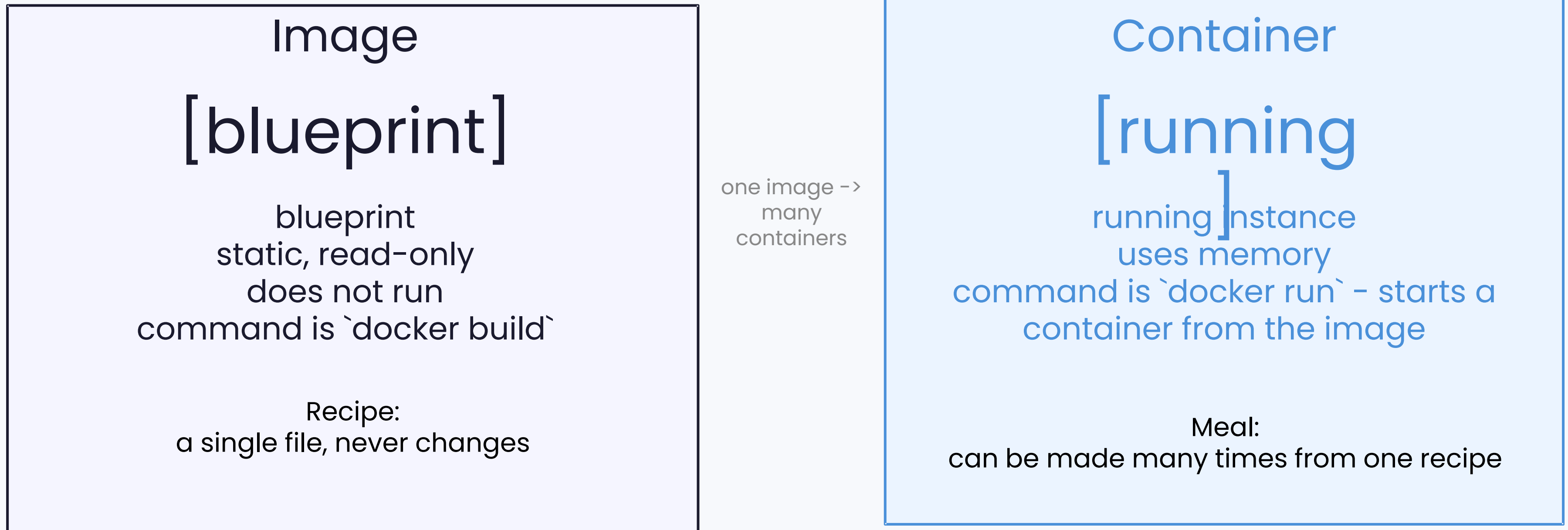


Docker Isolates More Than a venv



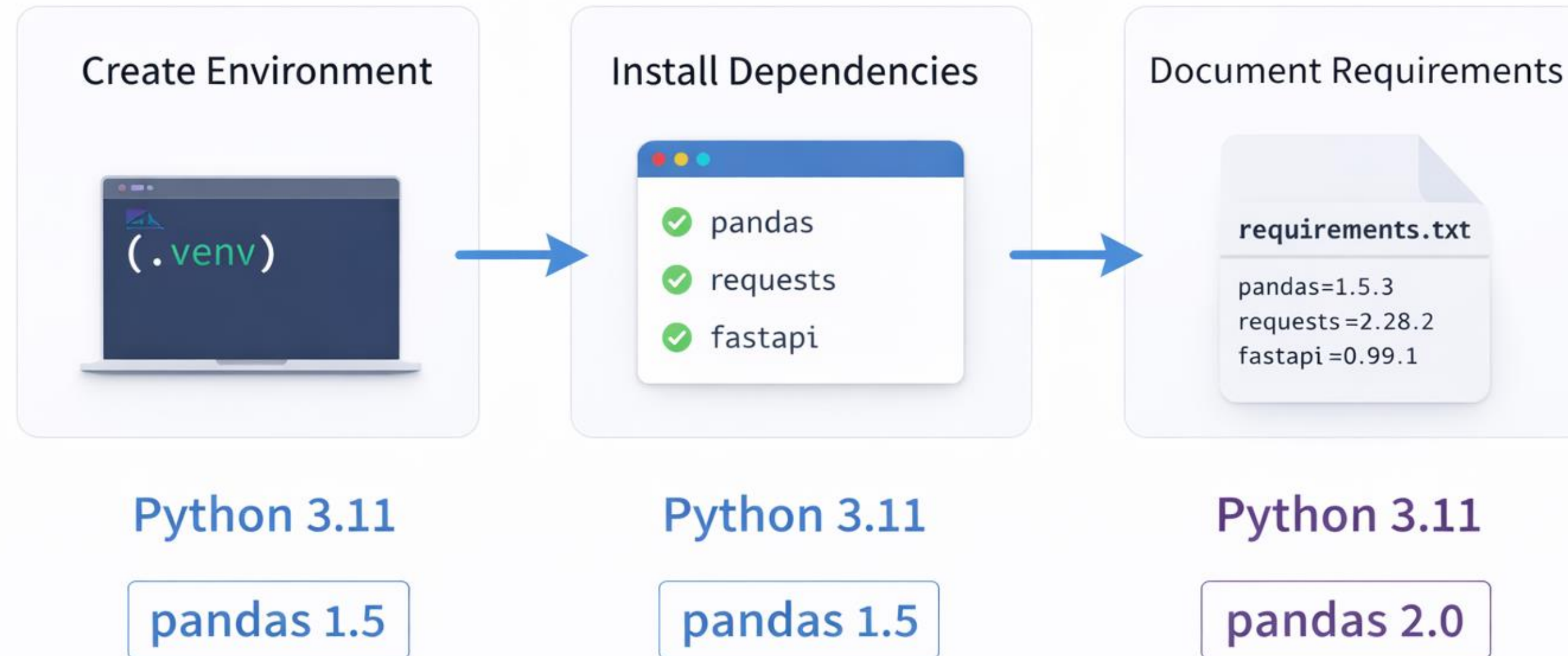
Same problem. Different scope.

Image vs. Container

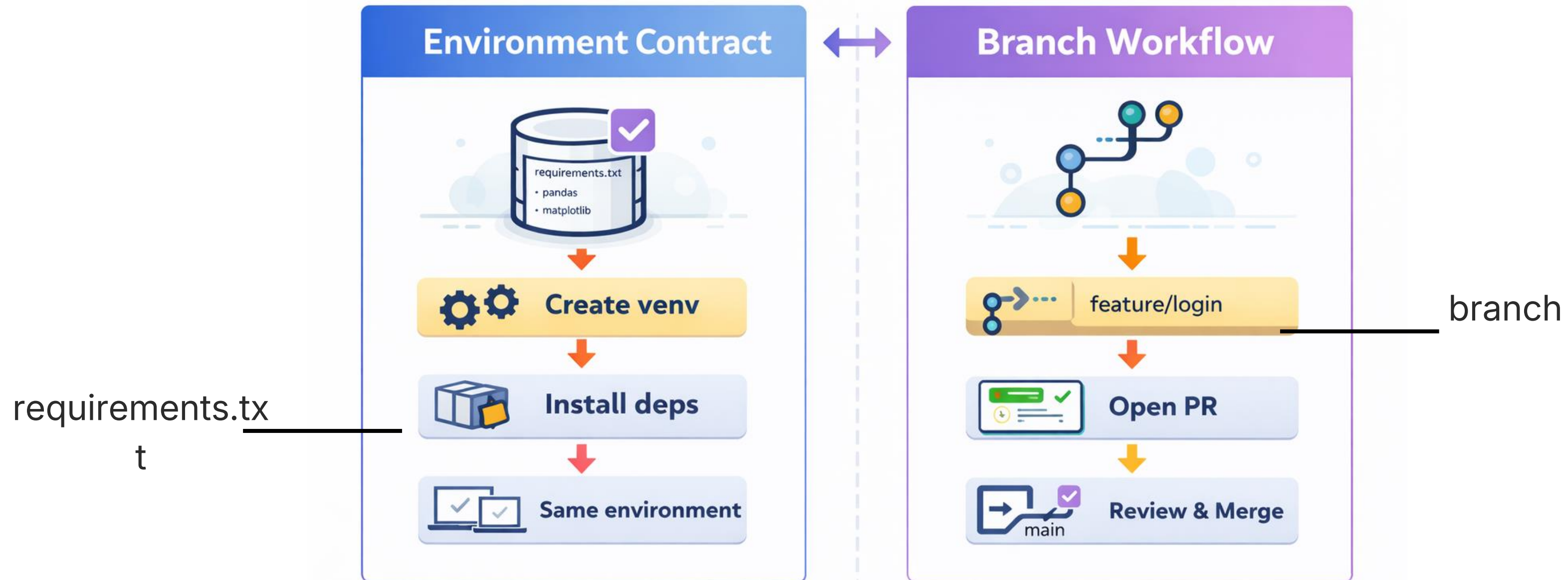


In later modules: you pull an image, then run a container from it.

ENVIRONMENTS AS CONTRACTS: REPRODUCIBLE ENVIRONMENTS



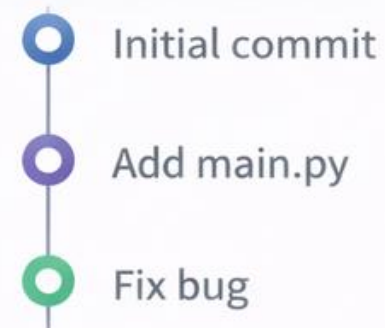
BRANCH WORKFLOW: GIT



A REPO IS MORE THAN A FOLDER WITH GIT



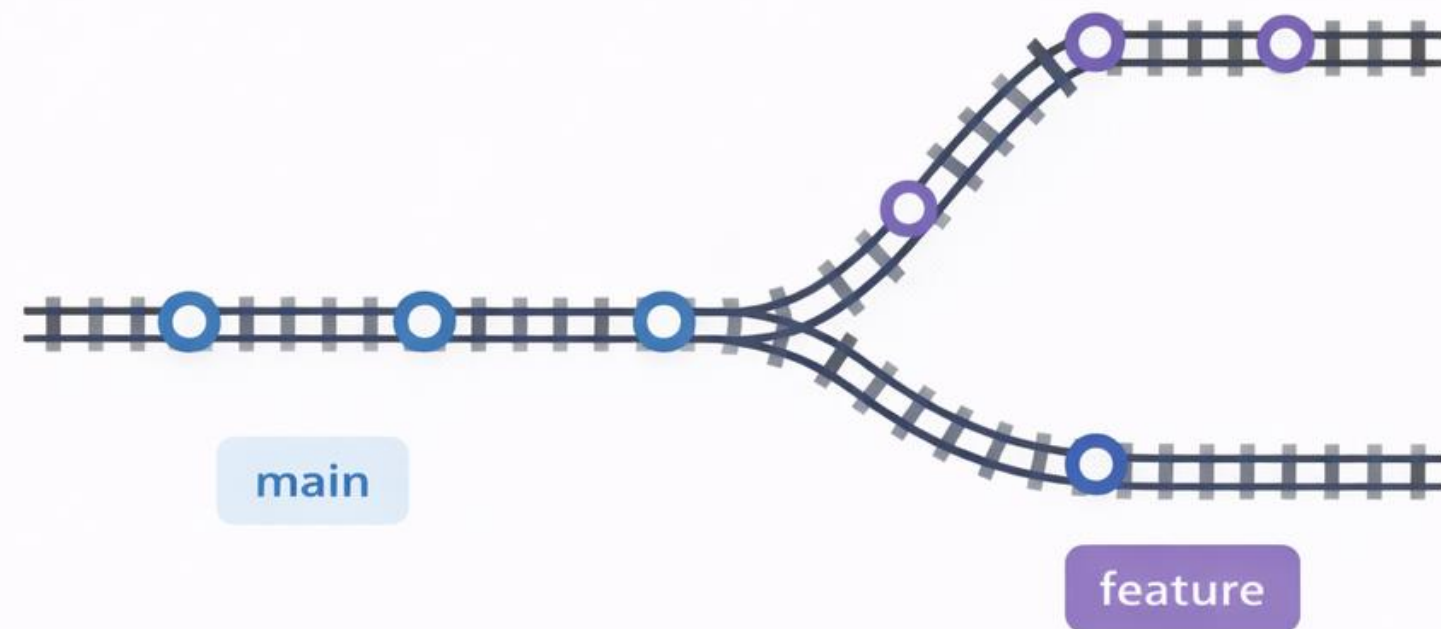
Folder



Git Repository

- Tracks changes
- Tracks authors
- Tracks history

PARALLEL WORK



- main → Existed / trusted code
- branch → new work

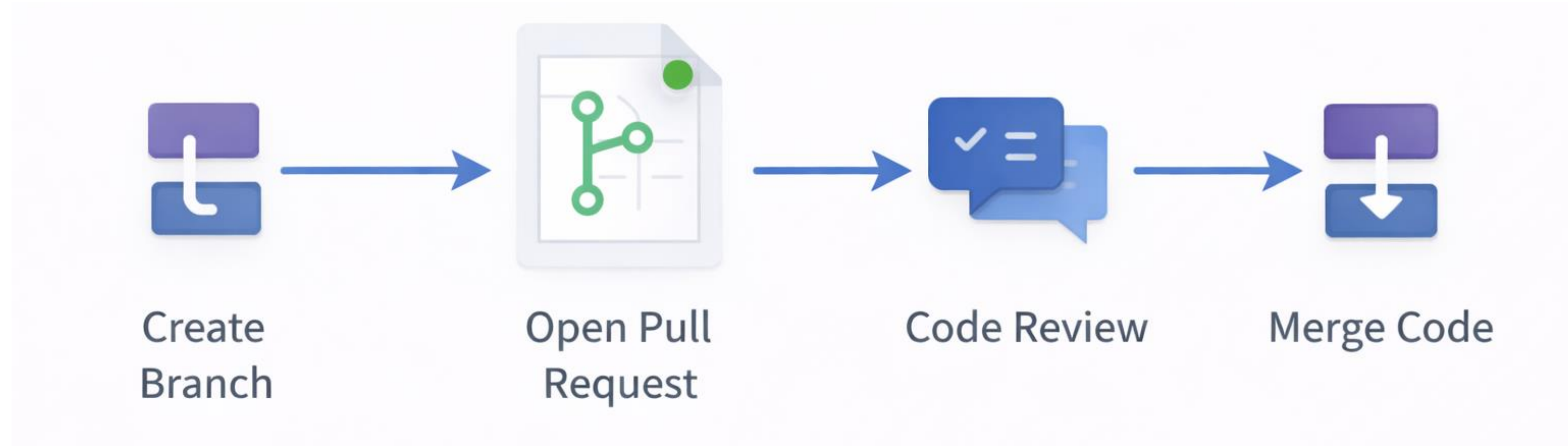
Your Local Repo and GitHub Are Two Copies



The repo on your laptop is a full copy — not a link to GitHub.

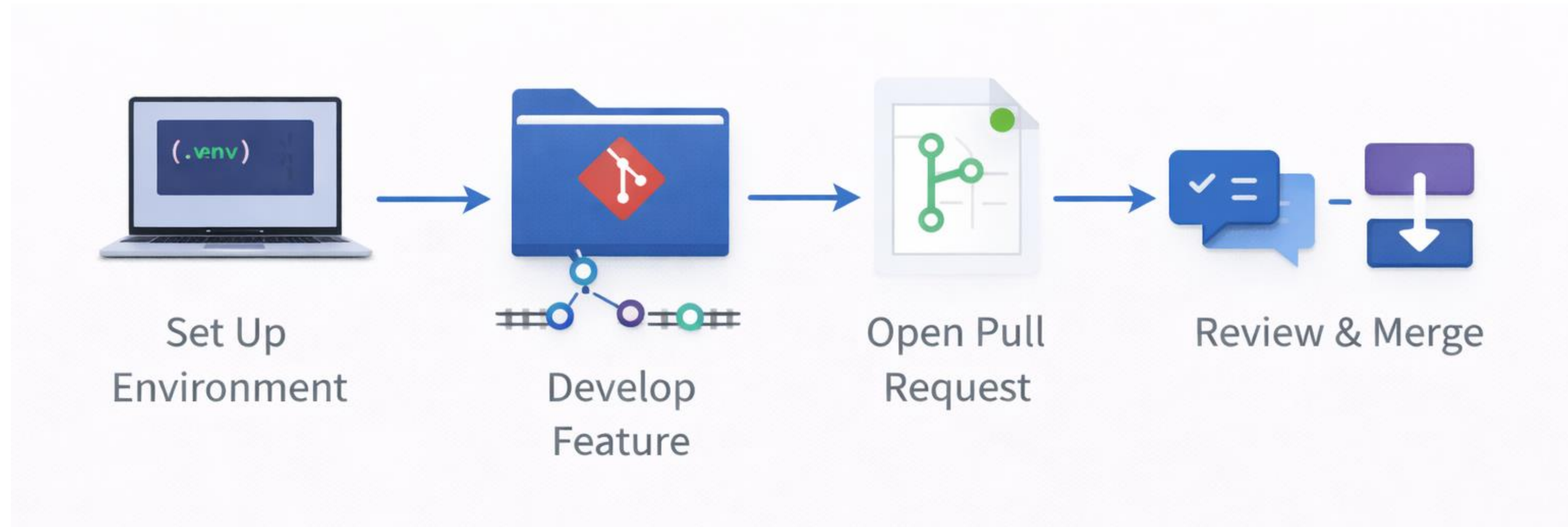
REVIEW BEFORE MERGE

BRANCH → PULL REQUEST → REVIEW → MERGE

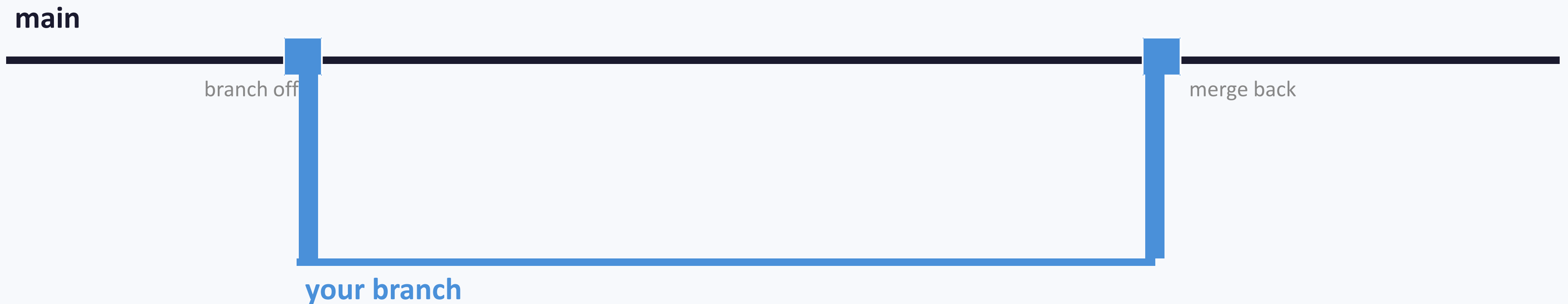


TEAM WORKFLOW

CREATE BRANCH → MAKE CHANGES → OPEN PULL REQUEST → REVIEW AND MERGE



A Branch Is Your Own Workspace



Work without touching what the team trusts.

A Branch Name Describes the Work

OK Good branch names

`setup/environment`

`fix/import-error`

`feature/data-pipeline`

X Bad branch names

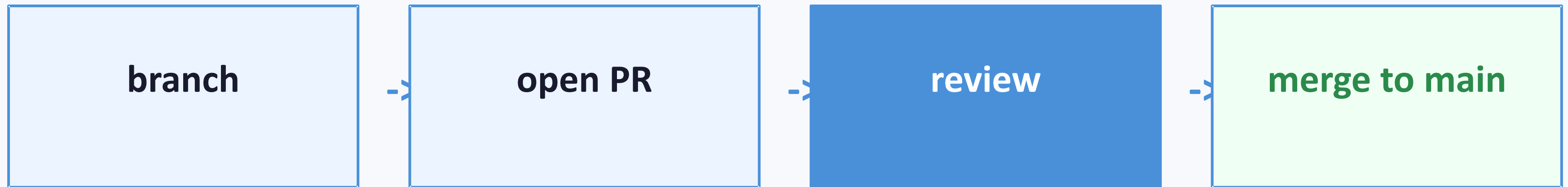
`test1`

`branch2`

`mybranch`

The autograder checks branch names. A wrong name = a failed test.

A Pull Request Asks: "Can My Work Join main?"



Pushing your branch **!=** merging to main.

A pull request is the review step in between.

The PR Description Is a Record of Your Decision

X Bad PR description

```
fixed stuff
```

(nothing else)

OK Good PR description

What changed

- Added requirements.txt
- Added .gitignore

Why

Team needs reproducible setup.

How to test

```
pip install -r requirements.txt
```

```
python test_environment.py
```

AI CODING ASSISTANTS

WORKFLOW: SUGGESTION → READ (HITL) → RUN → VERIFY → COMMIT



REMEMBER THESE TWO IDEAS



Your environment is a contract.

If someone cannot reproduce your environment, they cannot reproduce your code.



Git is a coordination protocol.

Branches and pull requests allow teams to safely build software together.